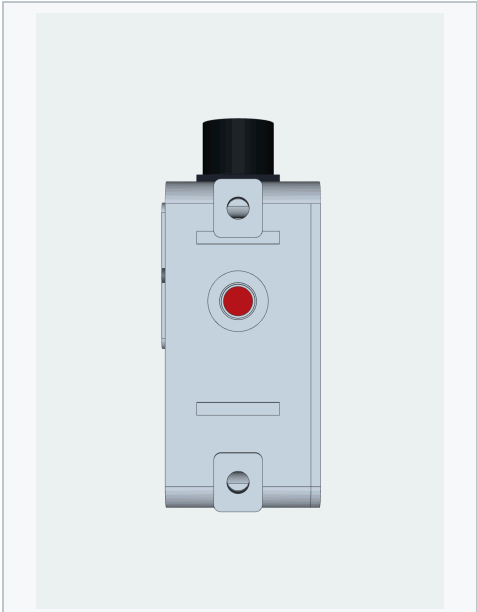


AETHER WATCH V1

Rugged handheld emergency-communications enclosure. Views are rendered from the validated parametric assembly; dimensions below are nominal millimetres.



FRONT · DISPLAY / STATUS



RIGHT · PROTECTED SOS



REAR · ACOUSTIC GRILLE

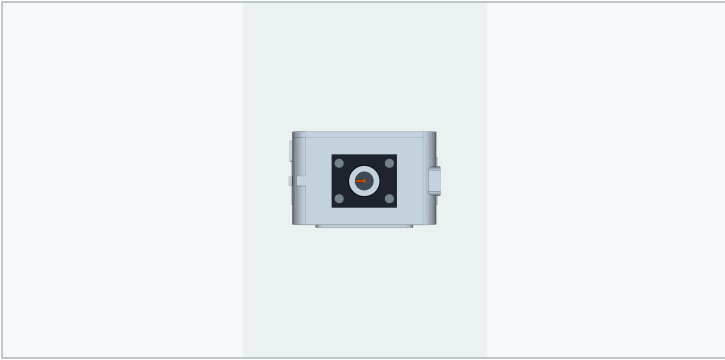
Nominal dimensions and feature callouts

FEATURE	MODEL DIMENSION / DETAIL
Overall assembled envelope	52.5 W × 33.2 D × 84.0 H, including antenna, button guard, bezel, and cover.
Main shell	50 W × 68 H × 30 D; 5 mm outer corner radius; 2 mm nominal front and side walls.
OLED nest / window	27 × 27 PCB pocket; 23 × 13 clear opening; raised bezel projects 1.2 mm.
SOS / status lamps	Ø7 side opening with guarded red plunger; 2 × Ø5.2 front LED bores at 12 mm centres.
USB / sound	10 × 5 bottom USB-C opening; rear grille is 7 × Ø2 holes over a 12 mm buzzer.
Antenna placeholder	Ø16 × 15 housing above the shell; Ø10.4 internal cable passage. RF hardware not specified.
Service cover	2 mm removable rear cover, four corner fasteners, fitted perimeter lip and screw-boss reliefs.

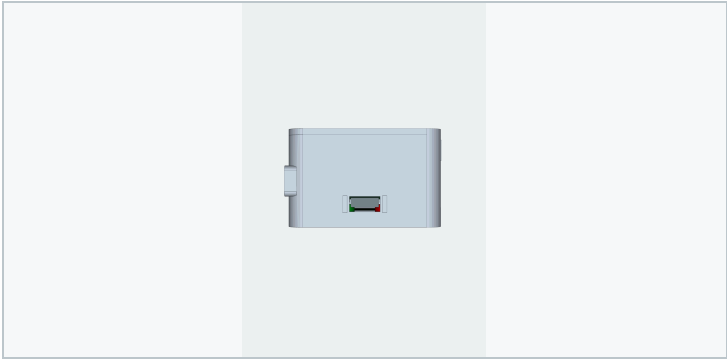
Envelope trade-off: the original 50 × 68 × 22 target cannot contain the specified 52 × 28 × 12 ESP32-C3 envelope, OLED, LiPo, buzzer, and clearances. The shell retains a 50 × 68 mm face; depth and total height grow to preserve a buildable stack and the 15 mm antenna.

Interface & internal stack

Top and bottom views show the antenna feed-through and centered USB-C access. The section diagram is schematic, derived from the assembled component envelopes.

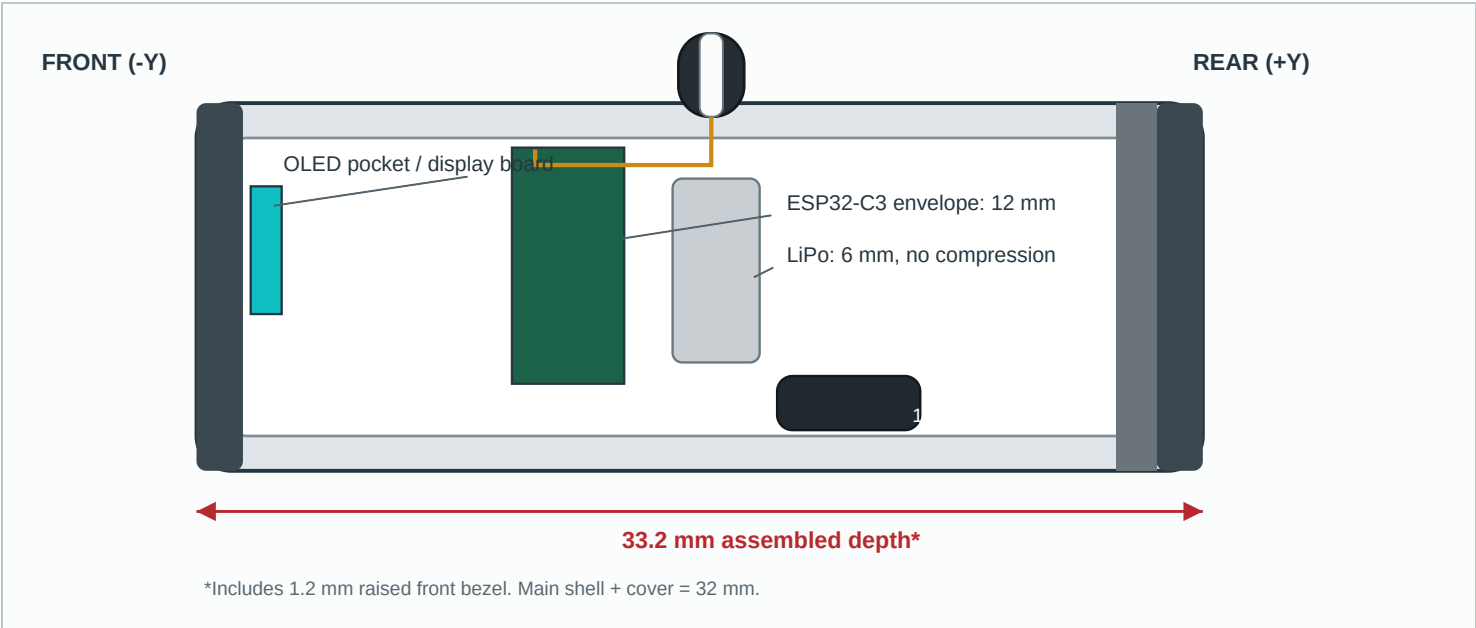


TOP · ANTENNA HOUSING



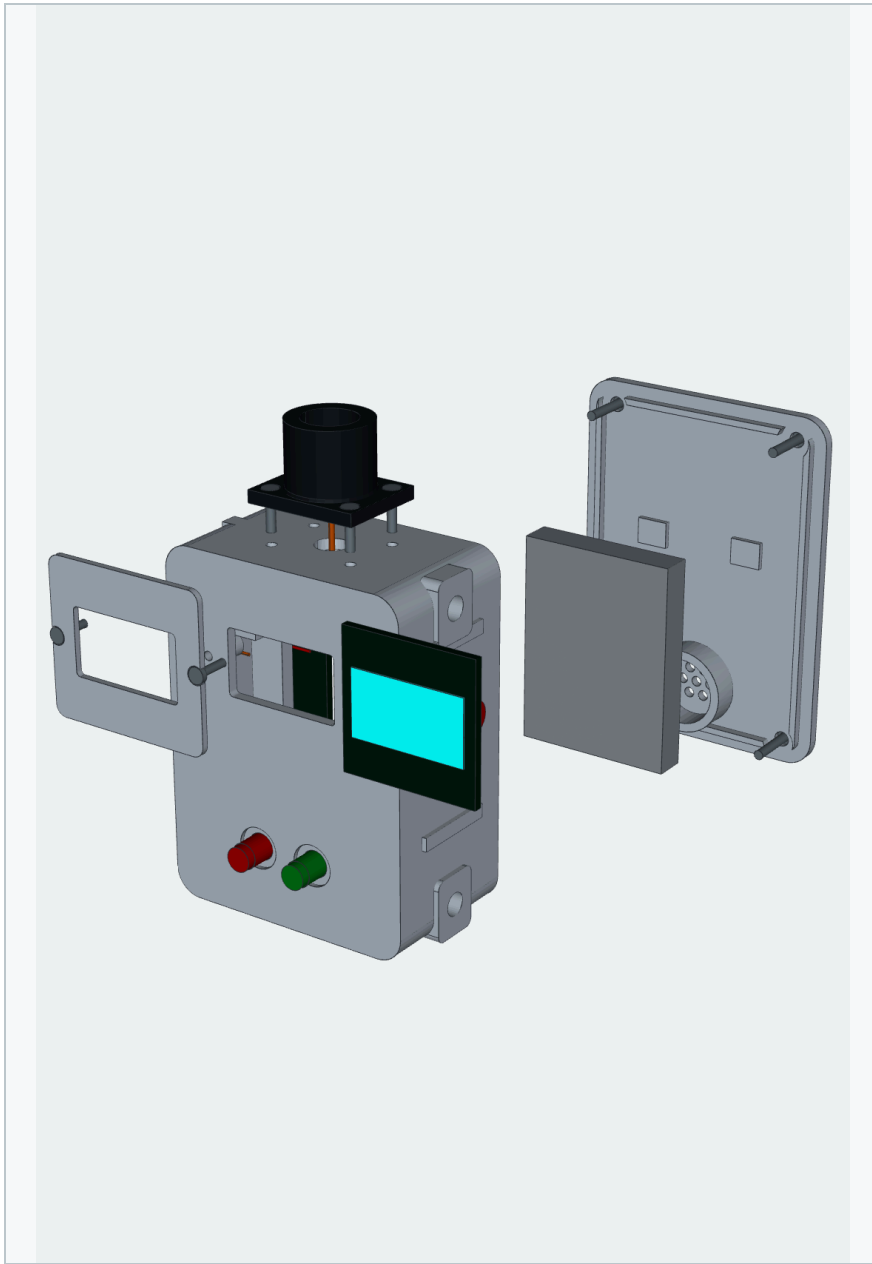
BOTTOM · USB-C PORT

Section A–A · front-to-rear depth stack (schematic)



INTERFACE	NOMINAL MODEL DETAIL
USB-C alignment	10 × 5 opening centred to generic ESP32-C3 short-end connector.
LiPo bay	32 × 41.2 clear opening around a 30 × 40 × 6 pack; 1.0 mm side and 0.6 mm end clearance.
Board mount	Four 4 × 4 support pads and removable M2 fasteners; generic 2.4 mm corner-hole pattern assumed.

Exploded assembly & build notes



EXPLODED ASSEMBLY · COMPONENT SPACING EXAGGERATED

Component key

- 1 Hollow antenna housing + coax route
- 2 Main front shell with display nest and side guards
- 3 Raised bezel and OLED PCB / active window
- 4 ESP32-C3 board envelope, supports and fasteners
- 5 3.7 V LiPo pack and paired lead route
- 6 Green / red 5 mm status LEDs
- 7 Guarded SOS plunger and 6 mm tactile switch
- 8 Rear 12 mm buzzer and seven-hole grille
- 9 Removable rear cover and four screws

Assembly order

1. Place OLED PCB in the 27.6 mm locating nest; fit bezel and two screws.
2. Seat LEDs; install the SOS tactile switch and side plunger.
3. Mount the ESP32-C3 on four supports; align USB-C opening.
4. Install battery in its locating tray; route leads in the side channel.
5. Seat buzzer on the cover cradle; route coax through the antenna passage.
6. Close rear cover and fasten four corner screws; attach strap.

FDM prototype notes

- PETG preferred; nominal 0.2 mm layer height, 0.4 mm nozzle.
- Print main housing front-face down; print cover outer face down with interior features upward.
- Inspect screw pilots and sliding lip after printing; tune for the actual printer/material.
- Electronics in the assembly STEP are fixed envelope solids, not print parts.

Prototype limitation: this is a mechanical enclosure model. The listed BOM has no GNSS receiver, satellite modem, or defined LiPo charge/protection circuit. It is not yet a functioning GPS or satellite-SOS communicator and is not for field deployment.

Design status: fit-checked generic envelopes; no production tolerances, IP rating, drop rating, RF tuning, electrical safety, or environmental qualification claimed. Verify the exact ESP32-C3 board and connector dimensions before printing.